Improved Object Detection on Forms using Synthetic Data and Self-Training

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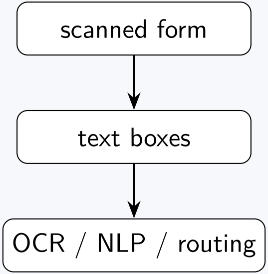
Bachelor Thesis Presentation

Research question: Can synthetic forms and self-training reduce additional manual box labels?

Motivation: forms need structure

- Forms appear in invoices, applications and administration.
- OCR gives words. Later steps often need regions.
- Manual bounding boxes are slow to create.



This work is a preprocessing step for document understanding.

Task: text blocks, not field semantics

Detection target

- one class: `text`
- one box per text segment
- no name, date or address class

Output

Text regions for later OCR, NLP or layout analysis.

Challenge: supervision cost and domain shift

Supervision cost

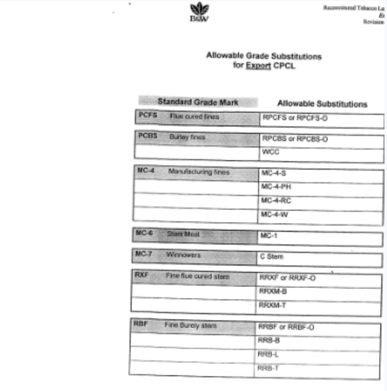
- FUNSD itself is fully annotated.
- Scaling to more real pages means drawing and checking many additional boxes.
- The goal is to reuse existing annotations instead of labeling every new real page.

Domain shift

- Synthetic pages preserve the document layout.
- Text appearance and background artifacts still differ from real scans.
- Self-training is used to adapt the detector to real pages.

FUNSD is small but fully annotated. The label problem here is the cost of adding more real training pages, not missing annotations inside FUNSD.

Data: FUNSD and how it is used here



Allowable Grade Substitutions
for Export CPCL

Standard Grade Mark		Allowable Substitutions
PCFS	Flue cured fines	RPCFS or RPCFS-O
PCBS	Burley fines	RPCBS or RPCBS-O
		WCC
MC-4	Manufacturing fines	MC-4-S
		MC-4-PH
		MC-4-RC
		MC-4-W
MC-6	Stem Meal	MC-1
MC-7	Winnowers	C Stem
RXF	Fine flue cured stem	RRXF or RRXF-O
		RRXM-B
		RRXM-T
RBF	Fine Burely stem	RRBF or RRBF-O
		RRB-B
		RRB-L
		RRB-T

FUNSD dataset

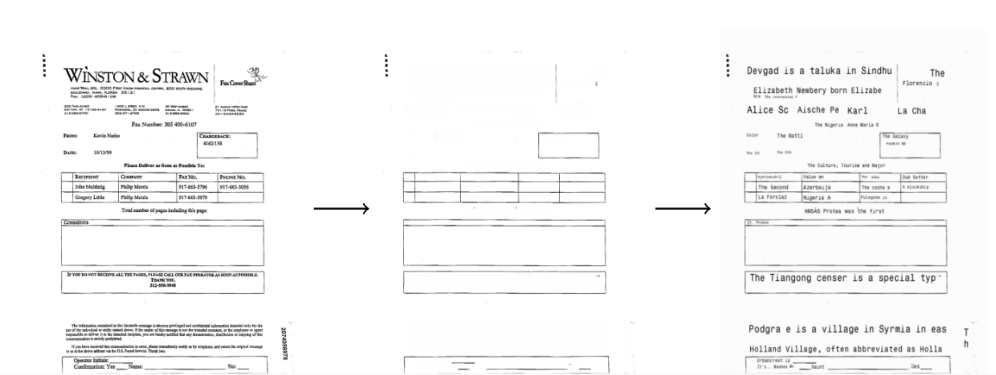
- 199 real, fully annotated scanned forms
- noisy scans with varied appearance
- 149 training, 50 testing forms
- word boxes and semantic entities

For this thesis

- single detection class: `text`
- synthetic pages reuse annotated layouts
- final evaluation stays on the real test split

Dataset facts: Jaume et al., FUNSD, ICDARW 2019. Most text is machine-written, with some handwriting such as signatures and dates.

Synthetic data generation



- Existing annotations mark the original text regions.
- Original text is removed while the document layout and non-text structure stay in place.
- New Wikipedia text is inserted and the resulting boxes are stored automatically.

Detector: Faster R-CNN

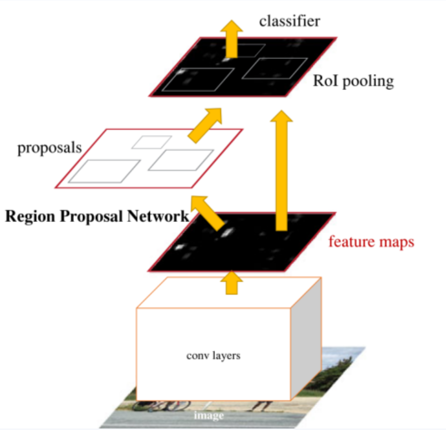
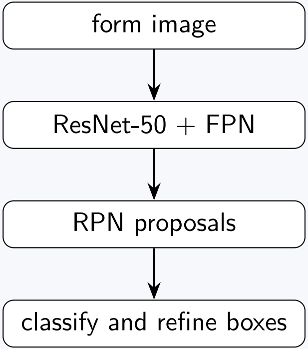


Figure adapted from Ren et al.



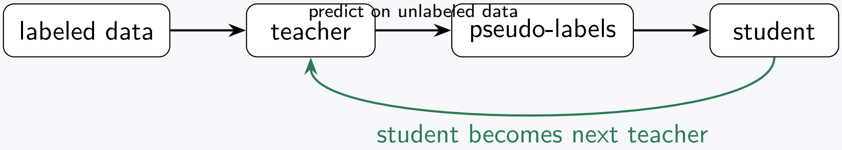
FPN provides features at several scales, which is useful for text blocks of different sizes.

Evaluation

prediction → ground truth → IoU ≥ 0.5

Metric	**Meaning**
Precision	fraction of predicted boxes that are correct
Recall	fraction of ground-truth boxes that are found
F1	harmonic mean of Precision and Recall

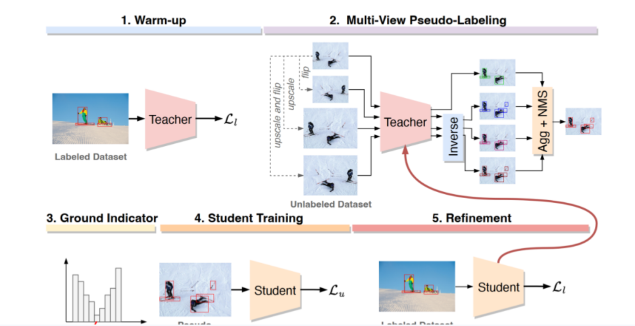
Self-training: teacher and student



- A teacher is trained from trusted labeled data.
- Confident predictions on unlabeled data are treated as temporary training labels.
- The student is trained with labeled and pseudo-labeled data, then the cycle can repeat.

The next slide shows how this general idea is adapted to the thesis.

ASTOD-inspired loop used in the thesis



ASTOD reference: Vandeghen et al., ICCV Workshops 2023.

- warm-up on synthetic pages
- teacher predicts boxes on real pages
- histogram threshold filters predictions
- student trains on synthetic labels and teacher-labeled real pages

Adaptation used here

Trusted synthetic labels remain in every round to limit error accumulation.

Experimental setup

Component	Setting
Detector	Faster R-CNN ResNet-50-FPN
Classes	background + text
Optimizer	SGD
Metrics	Precision, Recall, F1 at IoU 0.5
Initial teacher	trained on synthetic pages
Self-training	synthetic labels plus teacher-labeled real pages

The same detector, evaluation rule and data flow are used for the main comparisons.

Parameter choice: literature starting point, then our sweep

Experiment: Choose one stable optimizer setting before comparing the main experiments.

Literature starting point

- FUNSD evaluates Faster R-CNN as a text-detection baseline.
- Original Faster R-CNN training uses SGD, momentum 0.9 and weight decay 0.0005.
- ASTOD uses Faster R-CNN with FPN and a ResNet-50 backbone for teacher-student training.

Fixed setting in this thesis

Parameter	Value
Optimizer	SGD
Learning rate	0.006
Momentum	0.8
Weight decay	0.0005

Exact values were selected by preliminary sweeps on this setup, then kept fixed.

References: Ren et al., Faster R-CNN, 2017; Jaume et al., FUNSD, 2019; Vandeghen et al., ASTOD, 2023.

Baselines: size of the gap

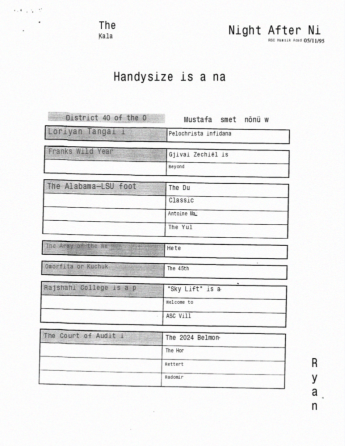
Experiment: How large is the gap before using self-training?

Training data	**Precision**	**Recall**	**F1**
Synthetic only	0.65	0.76	0.70
Real labels only	0.78	0.83	0.80

The synthetic-only model is about 10 F1 points below training with real labels.

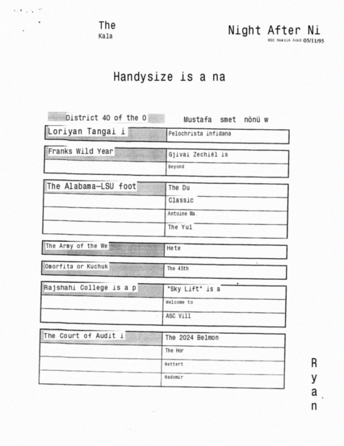
0.70 Synthetic only → **0.77** After self-training → **0.80** Real labels

Synthetic variants used in the comparison



Inpainting

Original text is removed. The surrounding scan background is preserved.



White background

Original text regions are replaced with white rectangles before inserting text.

Synthetic method comparison

Experiment: Does preserving the scan background improve transfer to real forms?

Method	Typical F1	Interpretation
Inpainting	about 0.76 to 0.77	preserves more real scan background
White background	about 0.75	simpler, but adds white-box artifacts

about 0.76 to 0.77
Inpainting

about 0.75
White background

Inpainting has a small advantage in the stronger comparison.

Amount of synthetic data

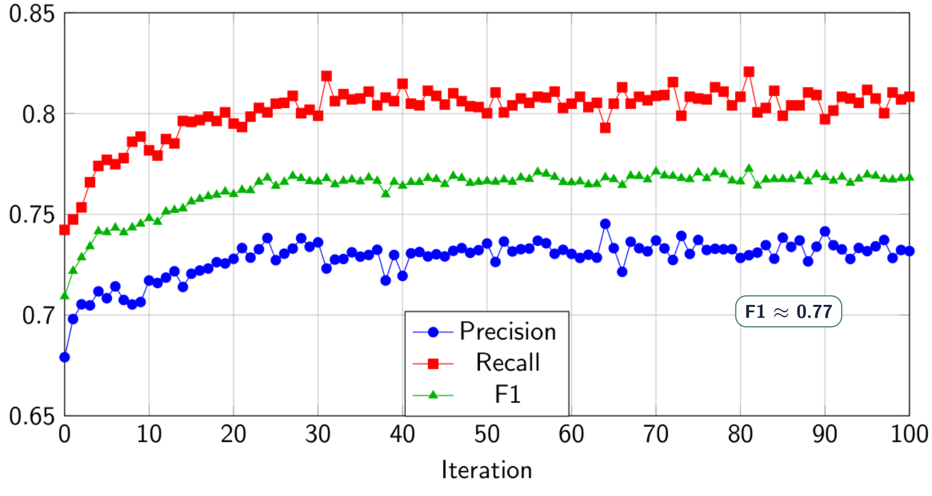
Experiment: How much synthetic data is needed for useful self-training?

Synthetic share	F1	Result
10%	about 0.67 to 0.72	useful, but limited
25%	about 0.72 to 0.75	already strong
50%	about 0.73	close to 75% in short runs
75%	about 0.73	little extra gain in short runs
100%	about 0.76 to 0.77	best long-run result

Result: reduced subsets still work, while the full long-run setting reaches the strongest result.

Main result: self-training over 100 iterations

Experiment: How does performance change as teacher-student iterations continue?



F1 rises from about 0.70 to about 0.77. Most of the gain appears in the first 20 iterations.

Ablation: keep trusted synthetic labels

Experiment: Do trusted synthetic labels prevent error accumulation during self-training?

Training in each round	Start F1	Later F1	Effect
Synthetic labels + teacher-labeled real pages	about 0.70	about 0.76 to 0.77	stable gain
Teacher-labeled real pages only	about 0.70	about 0.69	degradation

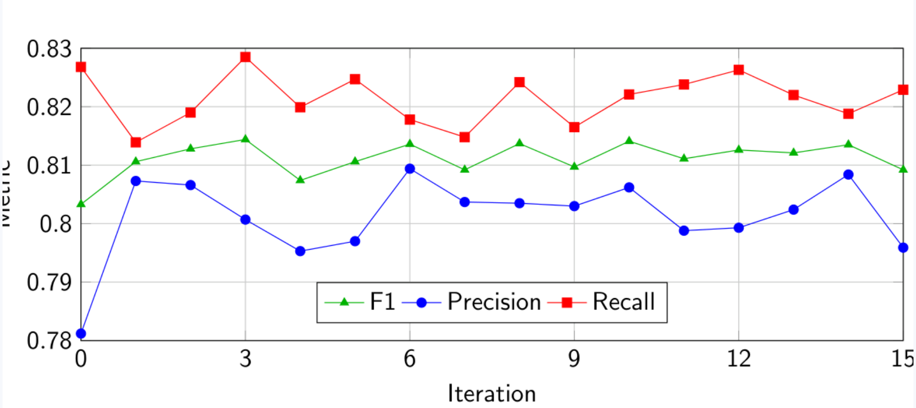
Pseudo-labeled real pages are real images whose training boxes were generated by the teacher model.

teacher error → used as training label → error can repeat

Keeping synthetic ground-truth labels stabilizes the loop.

Real-label experiments

Experiment: Is self-training still useful when the starting model already has real supervision?



0.801 real-only baseline

0.806 real-start self-training self-training

0.814 real + synthetic

The gain is smaller because the supervised starting model is already strong.

Main findings

1. Synthetic pages are a useful starting point: about 0.70 F1.
2. Self-training raises the result to about 0.77 F1.
3. Keeping synthetic labels prevents degradation from repeated pseudo-label errors.
4. Inpainting is slightly stronger than the white-background variant in the stronger comparison.

Real labels remain strongest, but self-training closes much of the gap.

Limitations

- Main evaluation on one form dataset.
- One detection class: `text`.
- Synthetic pages still have limited font and handwriting variation.
- More random seeds would make small differences easier to judge.

Small differences should be read as empirical trends.

Conclusion and outlook

- Synthetic forms plus self-training reduce the gap to supervised real-label training.
- The approach is most useful when additional real box labels are expensive.
- It does not fully replace real supervision.

Future work: more realistic synthetic pages, handwriting variation, transformer-based detectors, more datasets.

Thank you.

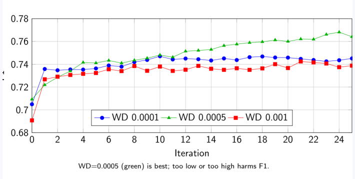
Backup: implementation details

Detail	Short answer
Dataloader	reads images plus annotation files
Pseudo-labels	teacher predictions saved as PT files
Adaptive threshold	21 bins in score range 0.5 to 1.0
NMS	reduces duplicate boxes before saving
Evaluation	confidence 0.5 and IoU 0.5

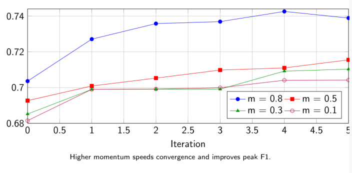
Backup: key numbers

Comparison	Value
Synthetic only	F1 about 0.70
Synthetic plus self-training	F1 about 0.76 to 0.77
Real-only supervised	F1 about 0.80
Real plus synthetic self-training	F1 about 0.814
Without synthetic anchor	F1 drops to about 0.69

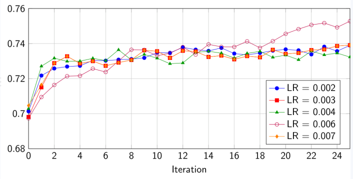
Backup: parameter sweeps



Weight decay



Momentum



Learning rate

These preliminary sweeps selected the values used on slide 12. They are shown here only if the parameter choice is questioned.